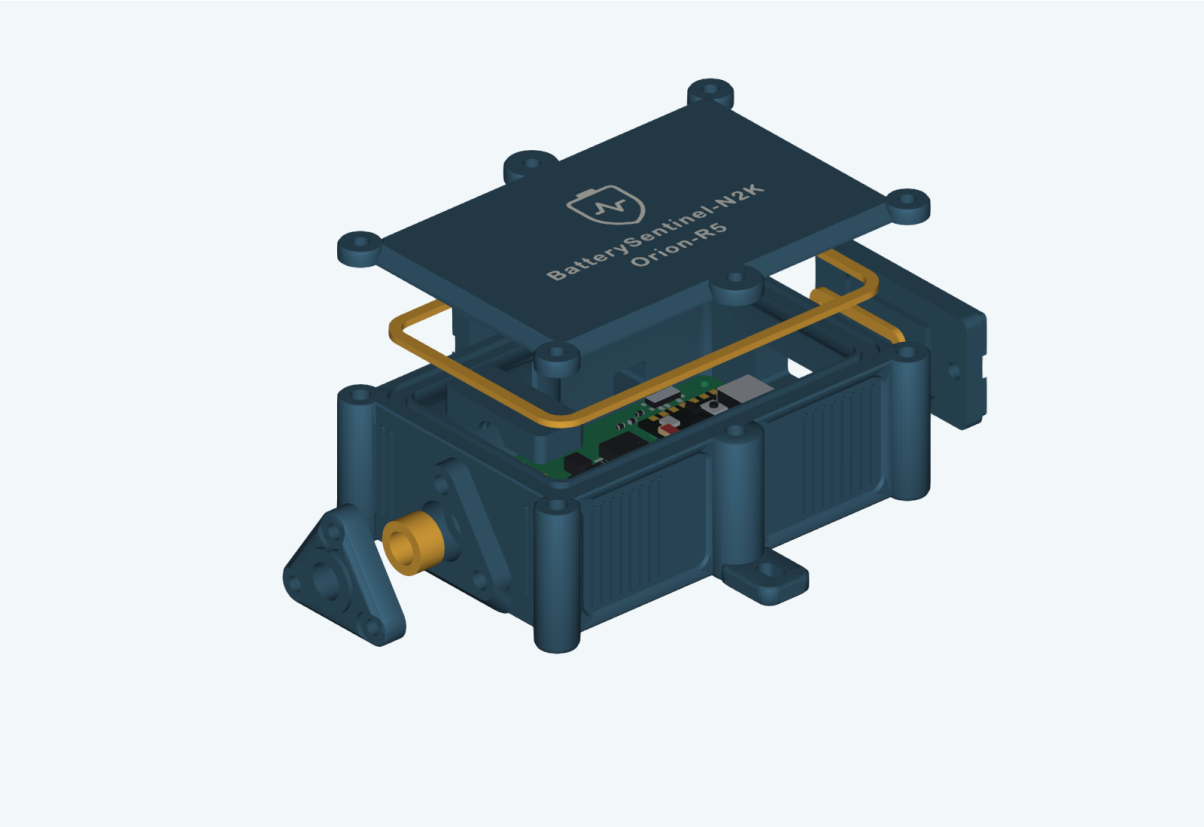
Enclosure assembly

V2.3 / 7 MM JACKET / SEPARATE STRAIN RELIEF



Actual V2 CAD. Amber: seals. Screws omitted; seated seal envelopes are illustrative, not a deformation simulation.

This manual applies to the current **V2.3** parts. The 88 x 57.2 x 33.2 mm main shell has 3 mm free wall fields, shallow ribs and six projecting screw bosses (96.8 x 68.8 mm envelope). Mounting tabs and covers extend farther. The sealing rim and connector pads are reinforced locally. The independent cable clamp, three-screw cap and centred flush artwork remain.

Prototype acceptance

The CAD and print checks are recorded in validation.json and package-validation.json. Physical fit, insert retention, clamp force, creep and water sealing remain to be qualified. There is no tested IP rating.

Before assembly, follow the separate **Manufacturing and TPU Air calibration** manual. Use coupons before a complete case. Keep the previous printed parts separate: V2 face seals do not fit the old 1.0 mm grooves.

Parts and tools

01 / SELECT ONE COMPLETE REVISION

Part / material	For one enclosure
base / PETG	1; includes PCB posts, lower saddle and clamp bosses
lid_body + lid_markings / PETG	1 co-printed two-colour lid; keep the shared coordinates
usb_cover, service_switch_cover / PETG	1 each; raised exterior reinforcement frame
cable_pressure_cap / PETG	1; three M3 holes; flat annular contact face
strain_relief_upper / PETG	1; upper ribbed jacket half-shell
clamp_shims_0p2 or 0p4 / PETG	One matched pair; alternatives, not both sets
lid_gasket, usb_gasket, service_switch_gasket / TPU Air	1 each; continuous rounded bead
cable_seal_b6p8_L6p8 / TPU Air	1 initial choice; nine variants supplied for coupon selection

Calibration items are not enclosure components: entry_fit_coupon, face_stop_coupon_1p4 / 1p6, face_test_gasket, flow_wall_coupon and shore_comparison_pad. The sleeve filenames describe the **smallest bead bore** and **free axial length** in millimetres.

Tools: caliper, suitable insert installation tool, M2/M3 drivers, flat washers, magnifier, flat reference surface, dry absorbent paper, cable marker and controlled pull-test fixture. A Shore A durometer is needed to measure the stated hardness targets; finger feel is not a measurement.

Fasteners and insert trials

02 / ALL EXTERIOR INSERT PILOTS STAY BLIND

Location	Hardware / quantity	Nominal engagement
Lid	6 x M3 x 8 + 0.5 mm flat washer	4.3 mm into the pilot
USB + switch covers	4 x M3 x 10 + 0.5 mm flat washer	5.5 mm; washer seat is 4.0 mm thick
Cable pressure cap	3 x M3 x 10 + 0.5 mm flat washer	4.5 mm; washer seat is 5.0 mm thick
Internal clamp	2 x M3 x 10 + 0.5 mm flat washer	3.3 mm with 0.2 mm shim
PCB	4 x M2 x 5; head <= 4.0 mm	3.4 mm through 1.6 mm board
Heat-set inserts	15 x M3; 4 x M2	Nominal body lengths 5.7 / 4.0 mm
Boat mounting	2 suitable screws + washers	Choose for mounting surface; 3.6 mm slots

M3 pilots: nominal diameter 4.0 x depth 6.7 mm. M2 pilots: 3.2 x 4.6 mm. These pilot sizes are provisional for the purchased insert bodies; thread designation alone does not establish compatibility. Use extra inserts for trials. The cap needs one more M3 set than before; the clamp needs two more.

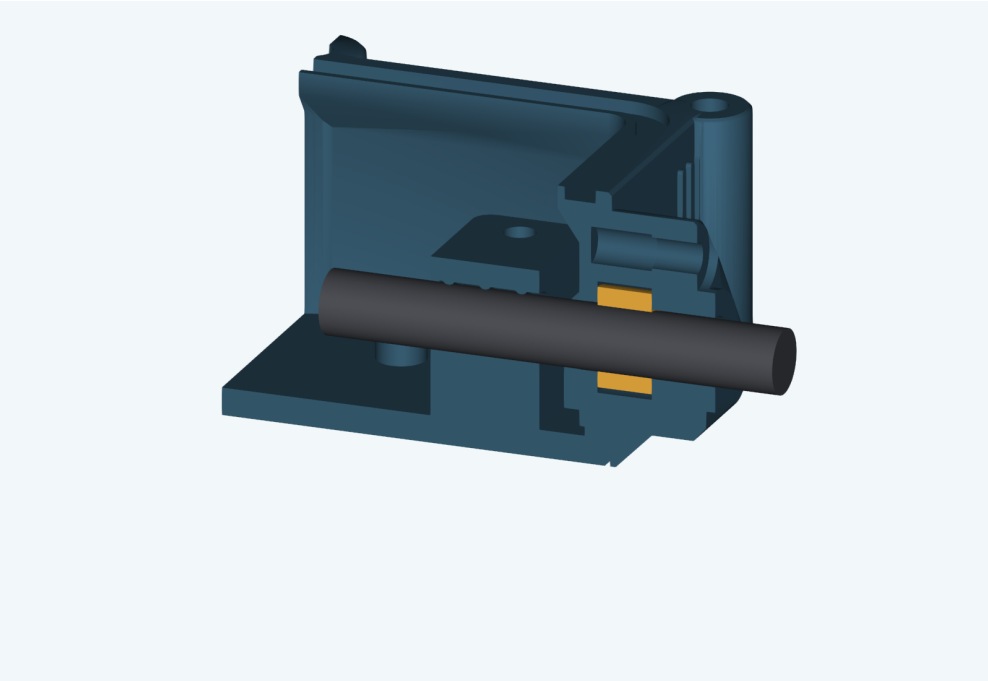
Check before heating

Measure the delivered inserts and screws. Validate insertion and resistance to tightening on the entry coupon. Install inserts flush and square, then allow them to cool. Do not drill a blind pilot through or bottom a screw in it. Use flat, not countersunk, heads. No tightening torque is qualified.

The engagement figures assume the stated washer thickness and fully seated inserts. Check actual lengths. A longer screw is not a remedy for a loose insert. Fasteners must suit the intended environment.

Cable seal installation

03 / THREAD THE BARE END BEFORE SOLDERING



CAD cutaway of the cable bay. The unsplit sleeve seals; the rigid clamp behind it takes cable load. Contact envelopes simplify TPU deformation.

1 **Disconnect all power**

Unplug NMEA power and USB. SERVICE alone does not disconnect all power. Keep the PCB out for wet testing.

2 **Thread the cable**

From outside to inside: pressure cap, unsplit TPU sleeve, enclosure wall. Feed the bare-wire end through; the connector does not pass through the small hole. Do not cut a slit into the sleeve.

3 **Seat the shoulder bead**

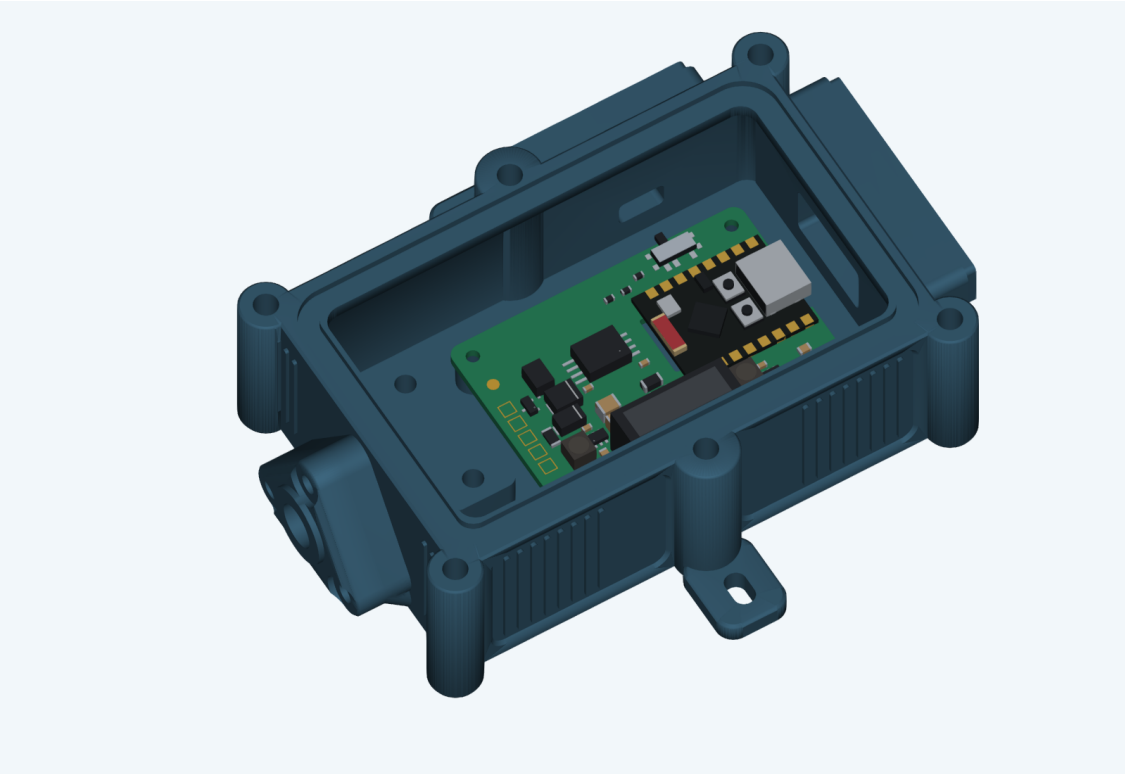
The sleeve end with the raised annular face bead points into the enclosure pocket, toward its internal shoulder. Keep the jacket intact through both the sleeve and the internal clamp.

4 **Close the three-screw cap**

Install the three screws and washers. Tighten in small alternating steps until the flat rigid lands meet the wall uniformly. Stop if the cap bows or an insert moves. Retention is adjusted at the separate clamp, not by overtightening the seal.

Internal strain relief

04 / CLAMP ONLY THE INTACT JACKET



Current PCB/component bodies and cable bay from CAD. Check the real solder tails, wire routing and driver access during the dry fit.

Place the jacket in the lower saddle. Fit a matched shim under each upper-clamp screw ear and position the ribbed upper half-shell. Start with the **0.4 mm** pair, then compare 0.2 mm and finally no shim only if retention is insufficient. Never use unequal shims.

Bring both screws down evenly. The bosses/shims limit travel; the shell must not bend. Smaller shims increase jacket compression. Choose the largest shim that prevents movement at the required service load without damaging the jacket. The bare nominal rib throat is approximately 6.7 mm; this is an adjustable trial geometry, not a universal cable clamp setting.

Mark the jacket at both sides of the clamp. On the coupon, test clamp retention independently with seal friction removed. Apply a controlled load representative of the installation and inspect movement, jacket damage and insert movement. Never pull on individual conductors or solder joints. Leave wire slack after the clamp and support the external cable near the enclosure.

If adjustment cannot give safe retention

Stop and revise the saddle or add a properly sized dense liner. Do not tighten beyond the rigid stops or crush the jacket. Recheck retention after 24-48 hours and after representative temperature exposure.

Face seals and closure

05 / BEAD ORIENTATION AND CONTROLLED COMPRESSION

Seal	Orientation	Closed geometry
Lid	Flat carrier in base groove; rounded bead faces lid	1.8 mm free -> 1.4 mm closed
USB cover	Flat carrier in cover groove; rounded bead faces case wall	1.8 mm free -> 1.4 mm closed
Switch cover	As USB cover	1.8 mm free -> 1.4 mm closed
Cable sleeve L6.8	Raised end-face bead toward internal shoulder	6.8 mm free -> 6.0 mm axial space

Remove strings, support debris and raised seams without nicking the sealing faces. Reject distorted covers or interrupted beads. Seat each loop continuously, without stretch, twist or adhesive lumps. Do not double-stack seals.

1 **Close service covers**

Fit each cover with two M3 screws and washers. Alternate small turns until its hard lands meet. The seal stays inside the screw holes; do not trap wires under a cover.

2 **Close the lid**

Fit six M3 screws loosely, then tighten alternating opposite locations in several passes until the flange lands meet. Keep wire slack entirely inside the seal loop. Stop if any part bows.

Do not mix V1 and V2 seals

V2 face grooves are 1.4 mm deep, with nominal 22.2% total-height squeeze. The old 1.2 mm flat seals and new 1.8 mm bead seals are not interchangeable. The 1.6 mm coupon is a comparison fixture, not the production groove depth.

Use the V2.3 base and lid as a matched set. Keep the sloped support below the lid rim intact. It stays below 45 degrees from vertical, including corners. Service planes and seals retain the styled coordinates (USB face 2.4 mm, switch face 3.4 mm outward from the original). Check plug reach and switch access. Open only when dry. Select SERVICE before USB; remove USB before RUN. Do not infer switch direction from the enclosure.

Empty-case acceptance

06 / BEFORE INSTALLING ELECTRONICS

1. **Dry fit**

   Check the actual PCB, modules, plug, screws and wire routing. Verify each insert pilot remains closed and each cap reaches its hard stops. Remove the electronics again before wet tests.

2. **Exposure test**

   Put dry absorbent paper inside, away from seals, and close the empty case with its real cable. Apply the intended splash exposure. No unverified immersion depth, pressure or IP rating is assumed.

3. **Inspect**

   Dry the exterior before opening. Inspect the paper, cable pocket shoulder, lid, service covers, insert areas and FDM walls. Any moisture is a failure. Locate the leak rather than merely increasing screw torque.

4. **Repeat under use conditions**

   After cable flexing, controlled pulling, 24-48 hours of compression and opening/closing cycles, repeat the leak test. Inspect jacket and clamp marks. Record exposure, load, duration and ambient temperature.

5. **Final electrical assembly**

   Dry all parts completely. Terminate the cable according to the current Orion Bridge wiring guide, confirm continuity/polarity, mount the board with four M2 screws and retain conductor slack.

6. **Commission closed**

   After successful mechanical acceptance, check power, temperature, Wi-Fi/Bluetooth and NMEA/Garmin operation with the final closed enclosure at the intended installation location.

Acceptance record

07 / RETAIN WITH THE ASSEMBLED DEVICE

Record	Result / date / initials
Cable OD / ovality; sleeve filename	_____
Insert body dimensions; screw lengths	_____
Lid / service / cable measured Shore A	_____
Filament batch; profile names; drying	_____
Clamp shim; applied pull load / duration	_____
Jacket movement / damage / insert movement	_____
24-48 h clamp and compression recheck	_____
Initial empty-case exposure / result	_____
Leak check after flexing and service cycles	_____
Electrical / thermal / RF acceptance	_____

If foam seeps or fails recovery, use a qualified conventional seal/gland and revise its seat. Do not release a failed printed seal on the basis of Shore hardness alone.

Source of geometry: parameters.json, mechanical-contract.json, Orion-Enclosure-V2.FCStd and validation.json in sealed-v2. Dimensions are nominal CAD dimensions; physical tolerances and load ratings are not established by the digital model.